

Element A: Presentation and Justification of the Problem

1. Use your team's design brief as a starting point. Make sure that the problem is explained in enough detail that *anyone* could understand the goal of this engineering design problem. Assume your reader knows nothing about the problem or its possible solutions. Include data and statistics* that help the reader understand the severity of the problem.
2. Use the problem statement as a starting point. Provide enough detail in your problem statement so that *measurable* design requirements may be established later (in Element C). Identify the stakeholders* and their role in the problem. Consider not only the obvious stakeholders, but also think about who or what might be affected by your solution.
3. Use your research* to justify your problem statement in detail. Explain why it is important to solve this problem now.
4. Site at least 2 *current, credible* sources in [MLA format](#).

*These are the sources that you used to find data/statistics, the research to justify your problem statement, and the interview with the stakeholder(s).

Source 1:

<https://mcleanmirror.com/3665/community/dont-let-hallway-traffic-make-you-tardy/>

Source 2: <https://westwoodhorizon.com/2022/09/students-speak-on-hallway-holdups/>

Hallway traffic is blocking the halls, stopping students from getting to class on time. We will know if our solution works when the student tardiness rate declines. But if it increases, we're doing the wrong thing.

Who is having the problem? - Students and staff.

What is the problem? - Hallway traffic is making students late.

Why is the problem important to solve? - Students need to be on time to class.

Where is the problem occurring? - The hallways.

When is the problem occurring? - Passing periods.

Who are the stakeholders? - Students and staff

Why are they the stakeholders? - Students are struggling to get to class on time, teachers need as much time as possible, and staff need to navigate the halls.

If the problem isn't solved soon, students will be serving wrongful lunch detentions for being late to class, when they can't control being late because of hall traffic.